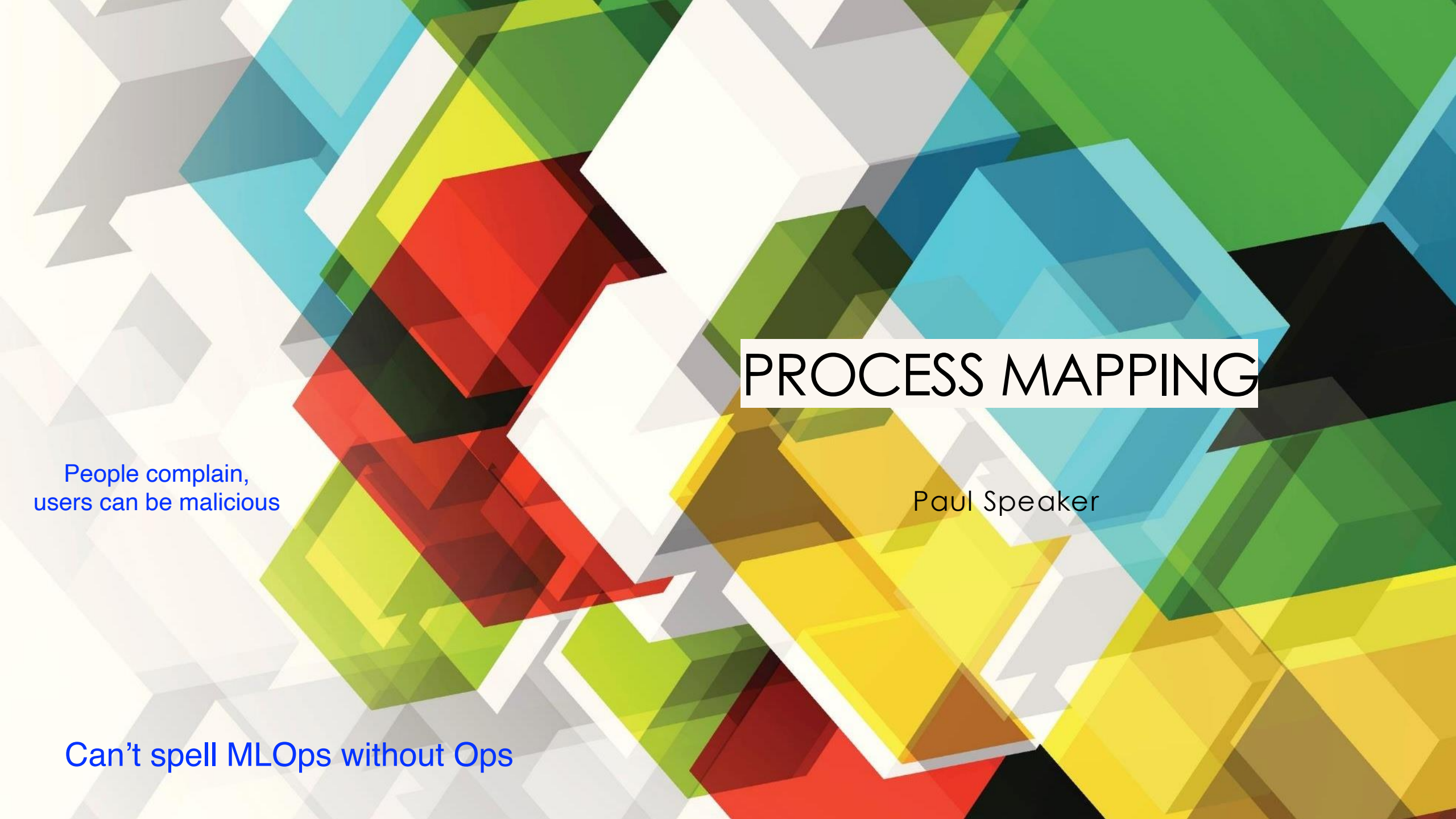




# PROCESS MAPPING

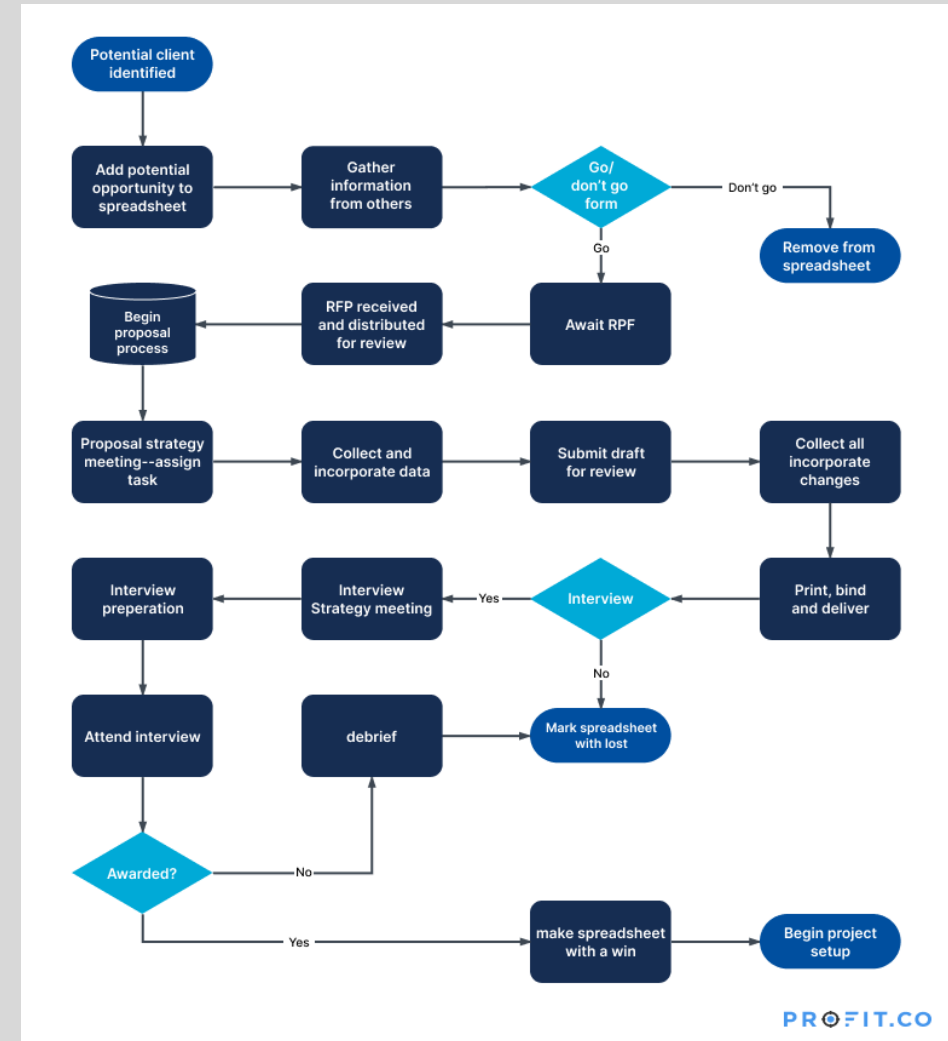
People complain,  
users can be malicious

Paul Speaker

Can't spell MLOps without Ops

# Process Mapping

- **Process mapping** is the visual representation of the workflow and tasks within a process.
- Why do we do process mapping?
  - Provides a clear overview of how processes function.
  - Helps pinpoint areas that can be streamlined or improved.
  - Identifies Bottlenecks and other possibilities of improvement



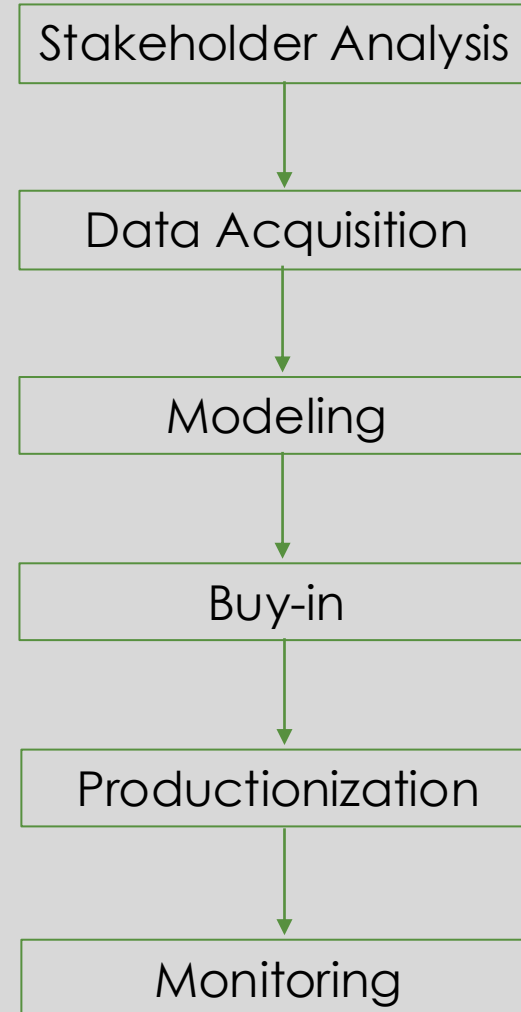
# Structure of a Process Mapping

- A Process mapping looks like a directed graph
  - Nodes represent tasks or decisions
  - Edges represent dependencies
- What does a good process map look like
  - Detailed enough to
    - identify where uncertainties are
    - Identify the bottlenecks
    - Show any differences between the project in question and what people may expect
  - Still simple enough to be intelligible

Simple - in content or in jargon? E.g., broad overview or in layman's terms?

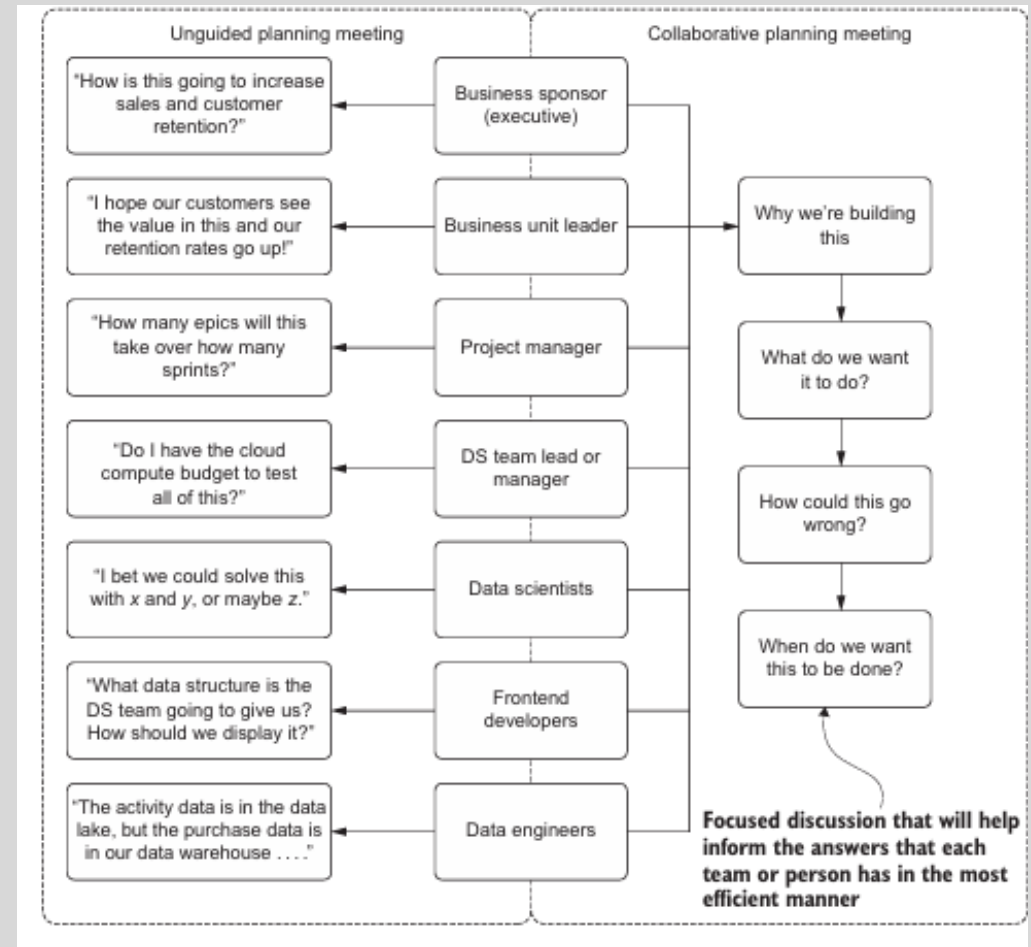
# Process Mapping for Data Science

- We will start with looking at a generic data science process
- What does this tell us? Not much! *Could be too simple?*
- Is there a data science project for which this is sufficient?



# Getting Information for a Process Mapping

- In context of process mapping, why are some questions good and some questions not helpful?



# Example 1: Fraud Detection

- As-is process: Transactions are processed using existing rules, customer history, and manual fraud review.
- Workflow fit: High-risk transactions may trigger a hold, authentication step, or analyst review.
  - target should be tied to something that you want to minimize
  - e.g., minimize time here. Includes time of response!
- How does it depend on time?
  - often, want to minimize loss of a transaction
- Human override: Investigators can clear or escalate transactions regardless of the score.
  - small transactions: yes/no
  - 10 plane tickets bought by someone in Peru: call them immediately
- Trust and incentives: Too many false alarms can cause analysts to ignore alerts.
- Needed information: Analysts need the score plus transaction context and reasons for the alert.

## Example 2: Hospital Re-Admission Risk

- What users do
- Before predictions: Clinicians identify high-risk patients using medical history and judgment. Or...there is no identification done at all
- Workflow fit: High-risk scores may trigger follow-up care or added discharge support.
- Human override: Clinicians can increase or reduce intervention based on other information.
- Trust and incentives: Clinicians may ignore a model they see as unreliable or opaque.
- Needed information: Users need the risk score, key drivers, and available interventions.

# What Game Theory Tells Us about Implementation

- Game theory models interactions between multiple agents as **games** of strategy between **players**.
- We will not go deeply into the theory of game theory, but several insights from game theory are useful to keep in mind
  - **Principal-agent problem:** model owner and user have different objectives
  - **Moral Hazard:** The person using the model is not the one bearing any negative consequences Predictive maintenance: will machines break down? If it breaks, big failure, need specialized team. "Somebody else's problem"
  - **Strategic Information Asymmetry:** Users are incentivized to provide false or incomplete information E.g., don't carefully follow outlined process and blame data scientist. Work with data subject of matter experts (SMEs). Many (20-25%) will willingly hide information, think being the only one who knows it is what gives them value
  - **Coordination Problems:** Users only benefit from a data science model if other people use it too frequent problem with supply chain: if predicting delays, need everybody working together (fails if one person not looking at results)
- How do these issues appear in Data Science solutions?

often overlap!